

HOMEWORK 12

6500-324: DATA MANAGEMENT FOR INFORMATION SYSTEMS (Fall 2022)

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Part 1. Multiple Choice or True/False

1. (1 point) What is CAP Theorem?
A) No distributed systems can be scalable as well as consistent
B) MongoDB cannot be scalable
C) A distributed system can meet only two of the following: consistency, availability, and errors tolerance
D) A relational database cannot meet the requirements of big data
2. (1 point) What is isolation property for a distributed system?
A) All transactions must be done sequentially one by one
B) All transactions should lead to the same result whether they are performed concurrently or sequentially
C) All transactions can be performed concurrently
D) All transactions must all succeed or fail as a whole
3. (1 point) Using JSON, each document is a list of key-value pairs. How do you enter a order data value like 2009-12-23?
A) OrderDate: new Date()
B) "OrderDate": ISODate("2009-12-23")
C) Order Date: new Date(2009, 12, 23)
D) OrderDate: ISODate(2009, 12, 23)
4. (1 point) How do you specify the value for for _id field of a document?
A) _id: 12345
B) "_id":ObjectId("12345")
C) _id:ObjectID(12345)
D) _id:"12345"
5. (1 point) How does MongoDB store a large document such as a photo or video file which is over 200MB?
A) use GridFS
B) use replication
C) use sharding
D) cannot be stored because the file is over 16 MB limit
6. (1 point) The following command will grant roles to scott. Which database does the user have readWrite permission?

```
<pre>
db.grantRolesToUser(
  "scott",
  [ "readWrite" , { role: "read", db: "reporting" } ]
)
</pre>
```

- A) reporting
B) the current database the user is in
 C) myDB
 D) admin
 E) not specified
7. (1 point) What is the command to insert the employee with eid 100, name "Scott" and hire date as 12/21/2019 into the collection employees?
 A) insert into employees (eid, name, hiredate) values (100, 'Scott', '21-Dec-2019')
 B) db.insert({eid:100, name:"Soctt", hiredate:ISODate("2019-12-21")})
 C) db.save({eid:100, name:"Soctt", hiredate:ISODate("2019-12-21")})
D) db.employees.insert({eid:100, name:"Soctt", hiredate:ISODate("2019-12-21")})
8. (1 point) The command db.students.find({same:"Lisa"}) will find all student documents with sname = "Lisa". How do you find the first document in the list of the found documents?
 A) db.students.findOne({same:"Lisa"})
 B) db.students.find({same:"Lisa"}).limit(1)
 C) db.students.find({same:"Lisa"}).sort(_id:1).limit(1)
D) all fo the above
 E) noen of the above
9. (1 point) How do you get the values of _id field for all the documents in students collection?
 A) db.students.find()._id
 B) db.students.findOne()._id
C) db.student.find().forEach(function(doc){printjson(doc._id)})
 D) db.student.find().forEach(function(doc){printjson(doc._id.getTimestamp())})
10. (1 point) (check all that apply) How do you show sname and bdate field vales for all the documents with sname = "Lisa"?
 A) select sname, bdate from students where sname='Lisa';
B) db.student.find({same:"Lisa"}).forEach(function(doc){printjson([doc.sname, doc.bdate])});
C) db.students.find({ sname "Lisa" }, { sname: 1, bdate: 1, _id: 0 })
 D) db.students.find({ sname "Lisa" }, { sname: 1, bdate: 1 })